

FYS4480/9480 September 17



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$$a_i |\Phi_0\rangle = |\Phi_i\rangle \quad |\Phi_0\rangle = |C\rangle$$

$i \in \{1, \dots, n-1\}$

$$|\Phi_0\rangle = a_1^+ a_2^+ \dots a_n^+ |0\rangle$$

$$a_n^+ |\Phi_0\rangle = 0$$

$$a_a^+ |\Phi_0\rangle = |\Phi^a\rangle \quad n+1$$

$$a \notin \{1, \dots, n\} \quad a_a |\Phi_0\rangle = 0$$

$$\overbrace{a_p a_q}^+ = \delta_{pq} \text{ if } p, q \in 1 \leq F$$

$$\overbrace{a_p a_q}^+ = \delta_{pq} \text{ if } p, q > F$$

$$a_a^\dagger a_n |\Phi_0\rangle = |\Phi_n^a\rangle$$

1p1k

$$a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle = |\Phi_{ij}^{ab}\rangle$$

2p2k

:

NpNk

∞ α

$N+2$ $\vdots$
 $N+1$ $\vdots$



$$\hat{H} = E_{Ref} + \hat{F}_N + \hat{V}_N$$

$$E_{ref} = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle = \sum_i \langle i | \hat{h}_0 | i \rangle + \frac{1}{2} \sum_{i,j} \langle ij | v | ij \rangle_{AS}$$

$$\hat{F}_N = \sum_{pq} \langle p | \hat{f} | q \rangle \{ a_p^\dagger a_q \}$$

$$\begin{aligned} \langle p | \hat{f} | q \rangle &= \langle p | \hat{h}_0 | q \rangle \\ &+ \sum_i \langle pi | v | qi \rangle_{AS} \end{aligned}$$

$$\hat{V}_N = \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \{ a_p^\dagger a_q^\dagger a_s a_r \}$$

$$\langle \Phi_n^a | \hat{H} | \Phi_0 \rangle =$$

$$\langle \Phi_n^a | E_{ref} | \Phi_0 \rangle$$

≈ 0

$$+ \sum_{pq} \langle p | \hat{f} | q \rangle \langle \Phi_0 | \overbrace{a_n^\dagger a_q a_p^\dagger a_q} | \Phi_0 \rangle$$

$$+ \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle_{AS}$$

$$\times \langle \Phi_0 | \{a_n^\dagger a_q\} \{a_p^\dagger a_q^\dagger a_r a_s\} | \Phi_0 \rangle$$

$= 0$

$$= \langle a | f^\dagger | i \rangle$$

$$\langle \Phi_{ij}^a | \vec{H} | \Phi_0 \rangle = \langle a | v | i \rangle_{AS}$$

$$\langle \Phi_i^a | \vec{H} | \Phi_j^b \rangle =$$

$$\delta_{ij} \underbrace{f_{ab}}_{\langle a | f | b \rangle} - \delta_{ab} f_{ij}' +$$

$$\langle a | f | b \rangle$$

$$\underbrace{\langle a_j | v | i \rangle}_{TDA}$$

Example ground state

$$|\psi_0\rangle = \sum_{PH} c_H^P |\Phi_H^P\rangle$$

$$= \underbrace{c_0^0 |\Phi_0\rangle}_{\text{open}} + \sum_{ai} c_i^a |\Phi_i^a\rangle$$

$$+ \sum_{\substack{ab \\ ij'}} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \dots$$

NPATH

intermediate normalization

$$\langle \Phi_0 | \Phi_0 \rangle = 1$$

$$\langle \psi_0 | \psi_0 \rangle = 1$$

$$C_0^0 = C_0 = \langle \psi_0 | \Phi_0 \rangle = 1$$

$\Rightarrow$, correlation

$$|\psi_0\rangle = (1 + \hat{C}) |\Phi_0\rangle$$

$$\hat{C} = \sum_{a \neq 0} C_a a_a^\dagger a_0 + \sum_{\substack{a \neq 0 \\ r_1'}} C_{r_1' a} a_a^\dagger a_{r_1'} + \dots + N \text{ part}$$

$$E[\psi] = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0$$

$$|\psi\rangle = \sum_k c_k |\Phi_k\rangle$$

$$k = \{PH\}$$

$$E[\psi] = \frac{\sum_{k,l} c_k^* H_{kl} c_l}{\sum_k |c_k|^2}$$

$$H_{kl} = \langle \Phi_k | \hat{H} | \Phi_l \rangle$$

Standard approach is to minimize $\langle \psi | H | \psi \rangle$

subject to $\langle \psi | \psi \rangle = 1$

$$\mathcal{L}[c, c^*] = \langle \psi_0 | H | \psi_0 \rangle$$

$$(\psi \rightarrow \psi_0) - \lambda (\langle \psi_0 | \psi_0 \rangle - 1)$$

$$= \sum_{p \in H} \sum_{p' \in H} (c_H^p)^* \langle \psi_H^p | H | \psi_H^{p'} \rangle c_H^{p'}$$

$$- \lambda \left(\sum_{p \in H} |c_H^p|^2 - 1 \right)$$

$C \rightarrow C + \delta C$ and keep terms to first order

$$\delta \mathcal{L} = \sum_{pH} \sum_{p'H'} \left\{ \delta [C_H^p]^* \right.$$

$$\left. < \Phi_H^p | H | \Phi_{H'}^{p'} \right> C_{H'}^{p'} \right.$$

$$+ (C_H^p)^* < \Phi_H^p | H | \Phi_{H'}^{p'} \right> \delta C_H^p$$

$$+ \lambda \left[\delta [C_H^p]^* C_{H'}^{p'} + C_{H'}^{p'} \delta [C_H^p] \right]$$

$$\star \delta_{pp'} \delta_{HH'} = 0$$

$$\sum_{p'H'} \langle \Phi_H^p | \hat{H} | \Phi_{H'}^{p'} \rangle C_{H'}^{p'}$$

$$= \lambda C_H^p$$



for every
set of C_H^p

$$\lambda = E \Rightarrow$$

$$\hat{H} \cdot C = E \cdot C$$

↑
Hamiltonian matrix

	0p0h	1p1h	2p2h	3p3h	...	NpNh
0p0h	Eref	fai	<ab/ab>	0	0	- 0
1p1h	fia	x	x	x	0	- 0
2p2h	<ij/ia>	x	x	x	x	0 - 0
3p3h	0	x	x	x	x	x 0
4p4h	0	0	x	x		
1	1	0	0	x		
1	1	1	0	x		
1	1	1	1	0		
NpNh	0	1	1	1	1	

First now

$$\langle \Phi_0 | \Psi_0 \rangle = 1$$

$$\langle \Phi_0 | \widehat{H} | \Psi_0 \rangle = \langle \Phi_0 | \Psi_0 \rangle E_0 =$$

$$\langle \Phi_0 | H | \Phi_0 \rangle \rightarrow \sum_{a_i} C_a \underbrace{\langle \Phi_0 | H | \Phi_a \rangle}_{f_{a_i}}$$

" E_{ref}

$$+ \sum_{\substack{ab \\ i,j}} C_{ij}^{ab} \langle \Phi_0 | H | \Phi_{ij}^{ab} \rangle$$

$\langle ab | \nu | ij \rangle$

$$= E_0$$

$$E_0 - E_{ref} = \sum_{ai} C_i^a \langle a | f | i \rangle$$

$$+ \sum_{ab, i'j'} F_{ij}^{ab} \langle ab | r | i'j' \rangle A_{5-}$$

eigenvectors
from
diagonalization
of

$$H \cdot C_i = E_i \cdot C_i$$

~~H~~

2nd now include a state $|\Phi_n^a\rangle$

$$\langle \Phi_n^a | \hat{H} | \Phi_0 \rangle +$$

$$\sum_{j^b} \langle \Phi_n^a | \hat{H} | \Phi_{j^b} \rangle C_{j^b}$$

$$+ \sum_{j^b k^c} C_{j^b k^c}^{\dagger} \langle \Phi_n^a | \hat{H} | \Phi_{j^b k^c} \rangle$$

$$+ \sum_{j^b k^c l^d} C_{j^b k^c l^d}^{\dagger} \langle \Phi_n^a | \hat{H} | \Phi_{j^b k^c l^d} \rangle$$

$$= E_0 C_n^a$$

$$= f_{ai} + \sum_{jb} \langle \Phi_n^a | E_{ref} + \vec{F}_0 + \vec{V}_0 | \Phi_j^b \rangle \\ \times C_j^b$$

$$+ \sum_{jk} \langle \Phi_n^a | E_{ref} + \vec{F}_0 + \vec{V}_0 | \Phi_{jk}^{bc} \rangle \\ \times C_{jk}^{bc}$$

$$+ \sum_{abc} \langle \Phi_n^a | E_{ref} + \vec{F}_0 + \vec{V}_0 | \Phi_{jke}^{bcde} \rangle \\ \times C_{jke}^{bcde}$$

$$\Delta E C_i^a = f_{ai} + \sum_b f_{ab} C_i^b$$

$$- \sum_j f_{ji} C_j^a$$

$$+ \sum_{jb} \langle a_j | v | i b \rangle_{AS} C_j^b$$

$$+ \sum_{jb} f_{jb} C_{ij}^{ab} +$$

$$\frac{1}{2} \sum_{jbc} \langle a_j | v | b c \rangle_{AS} C_{ij}^{cb}$$

$$- \frac{1}{2} \sum_{jkb} \langle j k | v | i b \rangle_{AS} C_{jk}^{ab} +$$

$$\frac{1}{4} \sum_{okbc} \langle j k | v | b c \rangle C_{ij}^{ab} C_{ok}^{bc}$$

Guest $C_n^a(0) = \frac{f_{ai'}}{\epsilon_{i'} - \epsilon_a}$

2nd-perturbation theory

Pairing Model

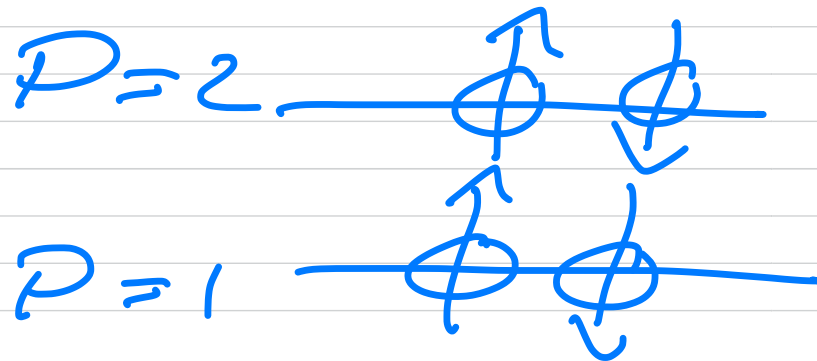
$n = 8$ (4 doubly deg
sp-levels)

$$N = 4$$

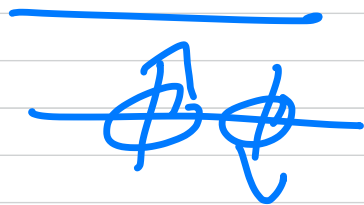
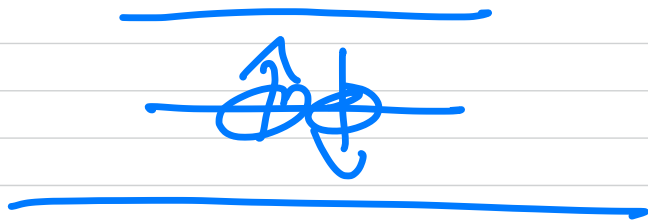
$D=4$ _____

$D=3$ _____

$$\#SD = \binom{8}{4} = 70$$



F
} $|\Phi_0\rangle$



2p_z



1p_z

$$\langle 0p_0 | H | 1p_1 \rangle = 0$$

$$\langle 0p_0 | H | 3p_3 \rangle = 0$$

$$\langle 0p_0 | H | 2p_2 \rangle \neq 0$$

$$\langle 0p_0 | H | 4p_4 \rangle = 0$$

no broken paint



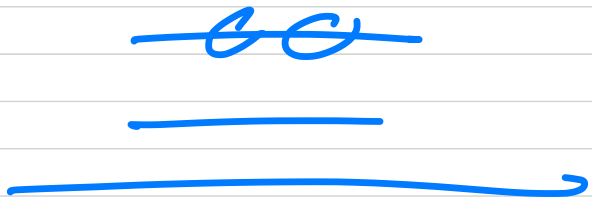
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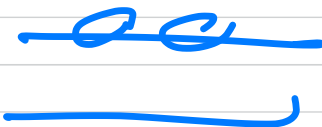
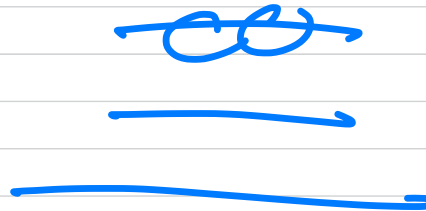
2p24



2 p 2 4



2 p 2 4



2p2u



CPZL